Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Carbon-14 is a radioactive substance that emits beta minus (β^-) particles.

(i) Which of these is the unit for the activity of a sample of carbon-14?

(1)

☐ **A** Pa
☐ **B** Nm
☐ **C** Gs
☐ **D** Bq

(ii) Which row of the table gives a correct comparison of the penetrating abilities of alpha, beta and gamma radiation?

(1)

	least penetrating ⟶ most penetrating
☐ **A**	alpha, beta, gamma
☐ **B**	alpha, gamma, beta
☐ **C**	gamma, beta, alpha
☐ **D**	gamma, alpha, beta

(iii) The mass of a beta minus particle is 9.1×10^{-31} kg.

Calculate the kinetic energy of a beta minus particle moving with a speed of 3.8×10^{6} m/s.

(2)

kinetic energy = ... J

(Total for question = 4 marks)

Q2.

Shot-put is an Olympic event.

The shot is a heavy ball.
An athlete throws the shot as far as possible.

A sports scientist analyses an athlete's throw to help improve performance.

In one throw, the shot continues to rise by another 1.3 m after it leaves the athlete's hand.
The mass of the shot is 7.26 kg.

Calculate the amount of gravitational potential energy gained by the shot.

(2)

gravitational potential energy gained = ... J

(Total for question = 2 marks)

Q3.

Figure 10 shows a skier sliding down a slope.

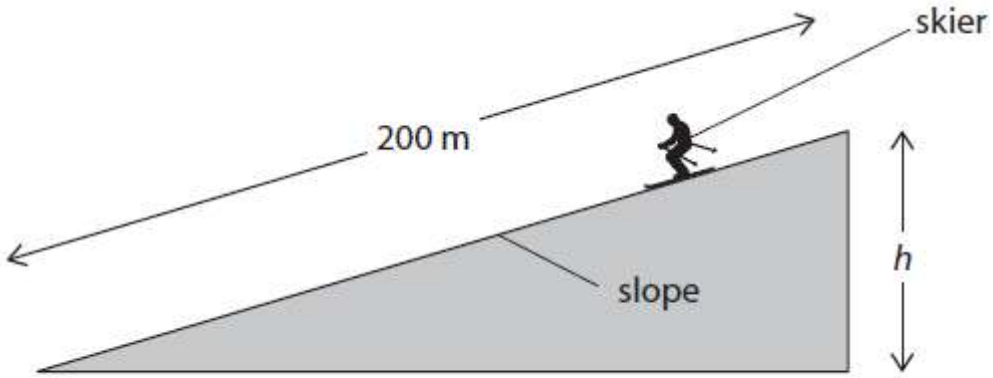

Figure 10

(i) The gravitational potential energy of the skier decreases by 39 000 J between the top and the bottom of the slope.

The mass of the skier is 70 kg.
Calculate the height, *h* of the slope.
The value of g is 10 N / kg.
Use the equation

$$\Delta GPE = m \times g \times h$$

(2)

h = ... m

(ii) The skier starts from rest at the top of the slope.

The velocity of the skier at the bottom of the slope is 27 m/s.
Using information from Figure 10, calculate the acceleration of the skier along the slope.

(3)

acceleration = ... m/s^2

(iii) When the skier slides from the top to the bottom of the slope, some energy is dissipated.

Explain how to determine the amount of energy dissipated.

(2)

.....

.....

.....

.....

(Total for question = 7 marks)

Q4.

Figure 13 shows two ice skaters during a performance.

Figure 13

(i) The two ice skaters are travelling together in a straight line at 3.50 m/s.

Their total momentum is 371 kgm/s.
The man has a mass of 64.5 kg.
Calculate the mass of the woman.

(4)

mass = ... kg

(ii) Calculate the kinetic energy of the man.

(2)

kinetic energy = ... J

(Total for question = 6 marks)

Q5.

A stone is held at rest above the ground.

The stone is released and falls until its velocity is 17 m / s.

Calculate the distance the stone has fallen when its velocity has reached 17 m / s.

(2)

distance = ... m

(Total for question = 2 marks)

Q6.

Figure 2 shows a person on a skateboard at the top of a ramp.

At P, the person is not moving.

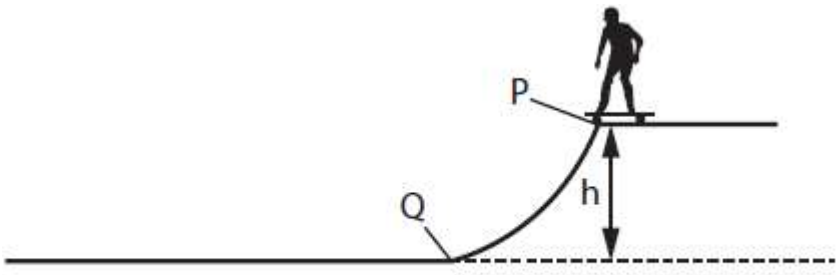

Figure 2

The person rides the skateboard down the ramp from P to Q.

The gravitational potential energy of the person decreases by 980 J.

The mass of the person is 35 kg.

Calculate h, the height of the ramp.

Use g = 10 N/kg.

Use the equation

change in gravitational potential energy = $m \times g \times h$

(2)

h = ... m

(Total for question = 2 marks)

Q7.

Figure 5 shows a football kicked against a wall.

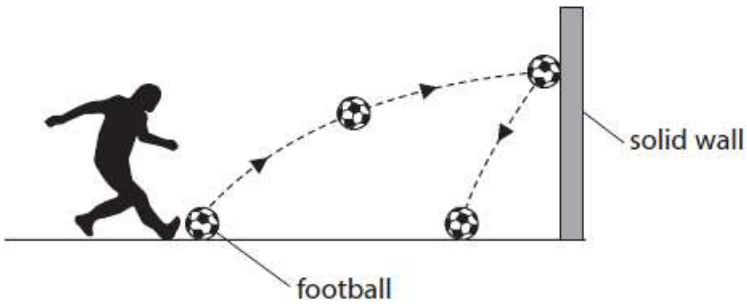

Figure 5

The football has a mass of 0.42 kg.

(i) The football gains 11 J of gravitational potential energy as it moves from the ground to the wall.
Calculate the height at which the ball hits the wall.

(3)

Gravitational field strength = 10 N / kg

Use the equation

$$\Delta GPE = m \times g \times \Delta h$$

height = ... m

(ii) Calculate the kinetic energy of the football when it is moving at a velocity of 12 m / s.

(2)

Use the equation

$$KE = \frac{1}{2} \times m \times v^2$$

kinetic energy = ... J

(iii) Describe the energy transfers that happen when the ball hits the wall.

(2)

.....

.....

.....

.....

(Total for question = 7 marks)

Q8.

(i) The aircraft lands with a velocity of 71 m/s.

The mass of the aircraft is 3.6 × 105 kg.

Calculate the kinetic energy of the aircraft as it lands.

(2)

kinetic energy of aircraft = ... J

(ii) When the aircraft has come to a stop, all the kinetic energy has been transferred to the surroundings.

Give **one** way that the energy has been transferred to the surroundings.

(1)

.....

.....

(Total for question = 3 marks)

Q9.

In a science fiction story, lightning is used as an energy source for accelerating a car.

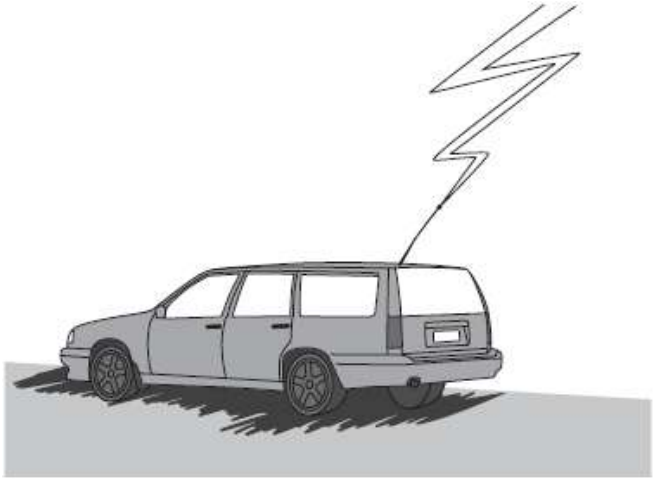

Figure 6

In the story, the car has a kinetic energy of 960 kJ at a speed of 40 m/s.

(i) Calculate the mass of the car.

(4)

mass = ... kg

(ii) Only 5% of the energy of the lightning bolt is transferred to the kinetic energy of the car.
Calculate the total energy of the lightning bolt in MJ.

(2)

energy = ... MJ

(Total for question = 6 marks)

Mark Scheme

Q1.

Question Number	Answer	Mark
(i)	**The only correct answer is D Bq** ***A*** *is not correct because it is the unit of pressure* ***B*** *is not correct because it is the unit of moment* ***C*** *is not correct because it is the unit of time*	**(1)** **AO1.1**

Question Number	Answer	Mark
(ii)	**The only correct answer is A** ***C and D*** *are not correct because they start with most penetrating* ***B*** *is not correct because gamma is more penetrating than beta*	**(1)** **AO1.1**

Question Number	Answer	Additional guidance	Mark
(iii)	selection of (KE =) ½ mv^2 (1) evaluation (1) 6.6×10^{-18} (J)	$\frac{1}{2} \times 9.1 \times 10^{-31} \times [3.8 \times 10^{6}]^2$ allow values that round to 6.6×10^{-18} (J) e.g. 6.5702×10^{-18} (J) award full marks for the correct answer without working	**(2)** **AO2.1**

Q2.

Question number	Answer	Additional guidance	Mark
	Substitution (1) PE = 7.26 x 10 x 1.3 Evaluation (1) 94.4 (J)		**(2)**

Q3.

Question number	Answer	Additional guidance	Mark
(i)	rearrangement (1) $h = \frac{\Delta GPE}{m \times g}$ evaluation (1) (h) = 56 (m)	$h = \frac{39\,000}{70 \times 10}$ allow numbers that round to 56 (m) e.g. 55.71 allow numbers that round to 57 (m) e.g. 56.85 (use of g=9.8 N/kg) award full marks for correct answer without working	**(2)** **AO3.2**

Question number	Answer	Additional guidance	Mark
(ii)	selection of $v^2 - u^2 = 2 \times a \times x$ (1) rearrangement (1) $a = \frac{v^2 (-u^2)}{2 \times x}$ evaluation (1) 1.8 (m/s^2) ALTERNATIVE METHOD calculation of time = 14.814 using time = distance / average speed (1) substitution (1) a (=(v-u) /t) = $\frac{27(-0)}{14.814}$ evaluation (1) 1.8 (m/s^2)	$27^2 (- 0^2) = 2 \times a \times 200$ allow $27^2 (- 0^2) = 2 \times a \times$ their (a)(i) $a = \frac{27^2(- 0^2)}{2 \times 200}$ allow $a = \frac{27^2(- 0^2)}{2 \times \text{their (a)(i)}}$ no ecf for evaluation allow numbers that round **down** to 1.8 (m/s^2) e.g. 1.8225 award full marks for correct answer without working.	**(3)** **AO3.2**

Question number	Answer	Additional guidance	Mark
(iii)	An explanation linking (measure) GPE at top/start and KE at bottom/end (1) energy (at top) subtract energy (at bottom) (1)	allow subtraction of any energies allow difference for subtraction award 2 marks for GPE at top subtract KE at bottom allow for 2 marks 39 000 – 25 515 OR 13485	(2) AO3.3a

Q4.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) 371 = (64.5+ m) x 3.5 rearrangement (1) m+64.5 = 371 / 3.5 evaluation of total mass (1) m+64.5 = 106 (kg) evaluation of woman's mass (1) m = 106-64.5 = 41.5 (kg)	full marks will be awarded for correct numerical answer without working	(4)

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) KE = ½ x 64.5 x 3.5^2 evaluation (1) 395 (J)	 allow answers which round to 395 e.g. 395.0625 full marks will be awarded for correct numerical answer without working	**(2)**

Q5.

Question number	Answer	Additional guidance	Mark
	selection **and** substitution (1) $17^{(2)}$ = 2 x 10 x distance evaluation (1) 14(.45) (m)	use of either $v^2 - u^2 = 2ax$ or $v^2 = 2gh$ $17^{(2)}$ = 2 x a x distance allow 289 for 17^2 award full marks for the correct answer without working give full credit for use of g=9.8 or 9.81 m/s^2 gives 14.7 (m)	**(2)** **AO2**

Q6.

Question number	Answer	Additional guidance	Mark
	Substitution or rearrangement (1) 980 = 35 x 10 x h	(Δh =) $\frac{\Delta GPE}{m \times g}$ or (Δh =) $\frac{980}{35 \times 10}$ allow use of 9.8 N/kg or 9.81 N/kg allow substitution into visible incorrectly rearranged algebraic expression	**2** **AO2.1**
	evaluation (1) (h =) 2.8 (m)	allow 2.85/2.86/2.9 for use of g = 9.8 N/kg or 9.81 N/kg ignore use of negative for decrease in GPE award full marks for the correct answer without working	

Q7.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) 11 = 0.42 x 10 x Δh rearrangement (1) (Δh =) $\frac{11}{0.42 \times 10}$ evaluation (1) 2.6 (m)	accept substitution and rearrangement in either order (Δh =) $\frac{\Delta GPE}{m \times g}$ accept any value which rounds to 2.6 (m) award 2 marks for 2.6 to any other power of 10 allow 1 mark for 0.38 allow 1 mark for 46(.2) award full marks for the correct answer with no working give full credit for use of g=9.8 or 9.81 N/kg (gives 2.7 (m))	(3) AO2

Question number	Answer	Additional guidance	Mark
(ii)	substitution(1) (KE=)$\frac{1}{2}$ x 0.42 x 12^2 evaluation (1) 30 (J)	 allow 30.2(4) (J) award 1 mark for 30 240 (J) award 1 mark for 2.52 (J) award 1 mark for 60.5 (J) award full marks for the correct answer with no working	(2) AO2

Question number	Answer	Additional guidance	Mark
(iii)	A description including: KE/kinetic (energy store) (1) (transfers to) and **one** of: elastic (potential energy store) (1) OR thermal (energy of ball/wall/surroundings) (1) OR <u>dissipates</u> (to surroundings) (1)	allow mechanically / mechanical transfer ignore reference to gravitational potential energy allow heat for thermal allow sound in this context ignore reference to the ground	**(2)** **AO2**

Q8.

	Answer	Additional guidance	Mark
(i)	selection and substitution (1) (KE) = $\frac{1}{2} \times 3.6 \times 10^5 \times 71^2$ evaluation (1) 9.1×10^8 (J)	 accept 9.07×10^8 (J) accept 907 380 000 (J) award full marks for correct answer without working do not award a power of ten error	**(2)** **AO2**

	Answer	Additional guidance	Mark
(ii)	any **one** from: mechanically (to the thermal store) (1) (heating) due to air resistance / friction (1) thermally (1)	 allow dissipated thermal (store) / heat (energy)	(1) AO2

Q9.

Question number	Answer	Additional guidance	Marks
(i)	recall K.E.= ½ m v^2(1) rearrangement (1) (m =) 2 x K.E. ÷ v^2 substitution (1) (m =) 2 x 960 000 ÷ 40^2 Evaluation (1) = 1200 (kg)	award full marks for the correct answer without working rearrangement and substitution in either order Ignore POT until evaluation	(4)

Question number	Answer	Additional guidance	Marks
(ii)	Use of efficiency equation (1) $\frac{960}{5}$ evaluation (1) =19 (MJ)	award full marks for correct numerical answer without working accept 19.2 (MJ)	(2)